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Ch.7 Structural and Computational Biology

Biology 11 • Punjab English Medium • PECTAA 2025–26 • Solved Exercise
Prompts are paraphrased and keyed to textbook question numbers. MCQ answer letters refer to the textbook options.

SECTION 1: MULTIPLE CHOICE QUESTIONS

Answer letters refer to the textbook options. Full textbook stems/options are not reproduced here.

1. Protein function generally depends on what?

Answer: (c) Three-dimensional structure — The folded 3D shape creates active sites and binding surfaces needed for function.

2. What are protein domains?

Answer: (a) Functional and structural units within a protein — Domains are independently folding regions that often carry out specific functions.

3. What is the first step in an X-ray crystallography experiment?

Answer: (d) Grow a crystal — A well-ordered protein crystal is needed before diffraction data can be collected.

4. What is the primary role of computational biology?

Answer: (a) Using computer algorithms to analyze biological data — It models, stores, compares and predicts biological structures and sequences.

5. Which approach predicts protein structure from amino-acid sequence using related structures?

Answer: (b) Homology modelling — Homology modelling builds a model based on an evolutionarily related template structure.

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SECTION 2: SHORT QUESTIONS

1. Define protein domains.

Domains are compact, independently folding units of a protein’s tertiary structure. Each domain often has a specific function, such as binding DNA, ATP or another protein.

2. How does coronavirus enter host cells?

The viral spike glycoprotein binds to a host receptor (e.g., ACE2 for SARS-CoV-2). Host proteases activate the spike, allowing membrane fusion or endocytosis, after which the viral RNA enters the cell.

3. Define genomics.

Genomics is the study of the complete genome of an organism, including sequencing, mapping, annotation and analysis of all genes and their interactions.

4. Differentiate between genomics and proteomics.

Genomics	Proteomics
Object: Whole genome/DNA.	**Object:** Whole proteome/proteins.
Focus: Genes, sequences, regulation.	**Focus:** Protein expression, structure, modification and function.
Methods: Sequencing, bioinformatics.	**Methods:** Mass spectrometry, 2D gels, protein arrays.

5. What is GenBank? Describe briefly.

GenBank is a public nucleotide sequence database maintained by NCBI. It stores annotated DNA/RNA sequences from many organisms and is used for sequence comparison, gene identification and evolutionary studies.

6. Write a short note on the Protein Data Bank.

The Protein Data Bank (PDB) is a public repository of experimentally determined 3D structures of proteins, nucleic acids and complexes, mainly from X-ray crystallography, NMR and cryo-EM. It is used for structural analysis, drug design and homology modelling.

SECTION 3: LONG QUESTIONS

1. Describe the applications of structural biology.

- **Determining active sites and domains:** 3D structures reveal the location, shape and environment of active sites and independently folding domains, e.g., polymerase and RNase H domains of HIV reverse transcriptase.
- **Identifying drug targets:** structures of disease-related proteins (e.g., SARS-CoV-2 spike protein) reveal pockets where drugs or vaccines can bind and block function.
- **Host–pathogen interactions:** structures show how pathogen proteins attach to host receptors and enter cells, guiding entry inhibitors.
- **Protein misfolding diseases:** structural studies explain how incorrect folding causes diseases such as cystic fibrosis, Parkinson’s and Alzheimer’s.
- **Protein engineering and drug design:** knowing structure allows rational modification of stability, activity and binding specificity.

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2. Write a note on the principle and working of X-ray crystallography.

Principle: when an X-ray beam strikes a crystal, atoms diffract the beam in specific directions. The angles and intensities of diffracted beams are used to calculate a 3D electron-density map, into which an atomic model is built.

> Protein purification → Protein crystallization → Expose crystal to X-rays at various angles → Record diffraction pattern on detector → Compute electron-density map → Build and refine atomic model

Crystals are needed because they arrange molecules in a regular repeating pattern. Crystallization may take hours to weeks and requires careful control of temperature, pH and salt concentration. The method gives high-resolution structures but cannot easily capture flexible or non-crystalline molecules.

3. Briefly describe key databases of structural and computational biology.

Database/tool	Content/use
GenBank	Public database of nucleotide sequences with bibliographic and biological annotations; used for genetic research and comparative genomics.
Protein Data Bank (PDB)	Repository of experimentally determined 3D structures of proteins, nucleic acids and complexes; used for structural analysis and drug design.
Ensembl	Genome browser providing genome sequences, gene models and comparative genomics for various species.
BLAST	Basic Local Alignment Search Tool; compares DNA/protein sequences to find homologues and predict function.
FASTA	Sequence alignment tool that searches databases for regions of similarity.

Together these resources store, retrieve and analyse biological sequences and structures, supporting genomics, proteomics and drug discovery.

INQUISITIVE QUESTIONS

1. How can structural biology help prevent a disease caused by a virus from a known family?

Knowing the family suggests likely structural features. Scientists can determine viral protein structures, identify conserved binding sites, design vaccines against surface antigens, and develop antiviral drugs that block entry or replication enzymes.

2. How can computational biology predict structure/function of an unknown protein from its sequence?

Use BLAST to find homologues, align sequences, identify conserved domains/motifs, build homology models if templates exist, predict secondary/tertiary structure, and infer function from conserved active sites and database annotations.

3. Comment on how homology models differ from experimentally determined structures.

Homology models are predictions based on related templates and may contain errors in loops, side chains or novel regions. Experimental structures (X-ray, NMR, cryo-EM) are directly observed and generally more reliable, though they also have resolution limits.

4. Draw a flowchart of drug development up to prescription.

> Target identification → Hit discovery → Lead optimization → Preclinical testing → Clinical trials Phase I–III → Regulatory approval → Manufacturing and prescription → Post-marketing surveillance

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